Devi Mutyala

dm53776@eid.utexas.edu | mutyala.devi17@gmail.com | mutyaladevi17.github.io | github.com/mutyaladevi17

Education

University of Texas at Austin **2021 – 2026**
Ph.D. in Mechanical Engineering, Oden Institute (GPA: 3.95/4.0) *Austin, TX*

Indian Institute of Technology, Delhi **2019**
M.S. (Research) in Mechanical Engineering (CGPA: 9.22/10) *New Delhi, India*

Indian Institute of Technology, Kanpur **2016**
B.Tech. in Aerospace Engineering (CGPA: 8.2/10) *Kanpur, India*

Technical Skills

AI/ML Frameworks: PyTorch, Hugging Face Transformers, Scikit-Learn, TensorFlow, OpenCV, torchvision
LLMs & Gen AI: OpenAI API (GPT-4, Whisper, TTS), Ollama, Google Gemma, Meta Llama, Gemini, LoRA / QLoRA, Fine-tuning, Prompt Engineering, Agentic AI, RAG
Data Science Tools: Gradio, BeautifulSoup, SQLite, MLOps (SageMaker), Jupyter, Git, Docker, Requests, lxml
Programming & DB: Python, MongoDB, ReactJS, SQL, R, MATLAB, Fortran, C/C++
Engineering Software: LAMMPS, COMSOL, ANSYS Fluent, SpaceClaim, PTC Creo, ParaView
Relevant Coursework: Fine Tuning Open Source LLMs, Deep Learning, Multimodal Systems, Machine Learning with Python, Data Science Lab, Software Engineering & Design, LLM Engineering

Professional Experience

Intel Corporation - Data Scientist (Internship) **June 2025 – Dec. 2025**
Design & Patterning AI Group (DPAI) *Santa Clara, CA*

- Engineered interpretable deep-learning defect detection architectures for semiconductor chip layouts; constructed an end-to-end **Computer Vision (CV)** pipeline capable of layout-to-image generation and model training.
- Developed robust ML workflows to address **severely limited labeled data** (~ 100 samples) and **zero non-defect data**.
- Designed and implemented a CNN-based **Prototypical Part Network (ProtoPNet)** to manage representative defect prototypes, optimizing for high coverage across defects. This enabled **explainable AI (XAI)** feature analysis.
- Integrated a modular, **end-to-end ML pipeline** encompassing data augmentation, model training, and results visualization into an existing defect commonality detection codebase.

AI & Machine Learning Projects

Medical Visual Question Answering (VQA) | *Gemma, LoRA, Hugging Face, Gemini, BioCLIP* **Feb. 2025 – May 2025**

- Built a multimodal Medical VQA system leveraging the **VQA-RAD dataset** to produce detailed clinical explanations for complex radiology images.
- Fine-tuned a generative text model (**Gemma-3-4B-it**) utilizing **QLoRA**. Engineered a custom descriptive dataset using **Gemini-2.0-Flash** to augment short-answer data for enhanced generative training (**BLEU 0.2265, ROUGE-1 F1: 0.48**).
- Developed benchmark classification pipelines utilizing **BioCLIP + BERT** and **DenseNet121 + BiLSTM**.

Multimodal Flight Assistant | *Agentic AI, OpenAI Tool Calling, Whisper, PyTorch, SQL*

- Architected a **multi-agent system** featuring a **dual-LLM orchestration architecture** to dynamically route open-ended dialogue and normalize geographic entities for complex flight booking workflows.
- Engineered an **Agentic AI** pipeline leveraging **OpenAI Tool Calling** (Function Calling) to securely map user intent to real-time **SQL** queries, enabling autonomous ticket price retrieval and dynamic updates.
- Built an end-to-end multi-modal interface integrating **OpenAI Whisper (STT)**, **TTS**, and **Gradio** for seamless voice-to-voice interactions.

Smart Audio Scribe | *OpenAI Whisper, Pyannote, Meta Llama, Hugging Face, PyTorch*

- Architected an end-to-end **Speech-to-Text (STT)** pipeline leveraging **OpenAI Whisper** for high-accuracy transcription and **Pyannote** for precise **speaker diarization** of multi-speaker audio.
- Engineered a **Generative AI** engine utilizing local **Meta Llama LLMs** via **Hugging Face Transformers** to autonomously parse transcripts and synthesize **context-aware meeting minutes** and action items.

AI Financial News Engine | *GPT-4o, GPT-5-nano, BeautifulSoup, Python, Requests*

- Engineered an intelligent financial data pipeline utilizing **GPT-5-nano** to autonomously classify and filter the most relevant market news from web-scraped (**BeautifulSoup**) Yahoo Finance pages.

- Integrated **GPT-4o** to synthesize extracted content into professional, structured **stock news summaries**, managing high-volume data extraction with robust error handling for real-time market insights.

Website Summarizer | *Ollama (Llama 3.1, Phi-3), OpenAI SDK, BeautifulSoup, Python*

- Developed a standalone, privacy-focused text extraction utility that leverages **local Ollama models** to perform serverless webpage summarization without external API calls.
- Implemented a robust parsing engine using **BeautifulSoup** and **lxml** to clean and strip noise (scripts/styles) from web content, delivering high-fidelity markdown-formatted summaries directly in notebooks.

Find Common Object (Explainable AI / ProtoPNet) | *PyTorch, torchvision, CNN, Matplotlib*

- Developed a CNN-based **Explainable AI** model implementing Prototypical Part Networks (**ProtoPNet**) to automatically discover and localize common sub-objects across image classes.
- Designed models that learn human-interpretable image prototypes rather than black-box embeddings, providing spatial localization through **bounding boxes, heatmaps,** and matched prototype visualizations.

Engineering & Research Experience

Oden Institute, University of Texas at Austin – Graduate Researcher **Sept. 2022 – May 2025**

Advisor: Prof. Narayana Aluru *Austin, TX*

- Investigating complex physics of nanoscale systems to explore the development of a **nanofluidic memristor**.
- Employing **Molecular Dynamics simulations (LAMMPS)** and **Poisson-Nernst-Planck continuum models (COMSOL)** to analyze ionic transport and electrokinetic behavior in nanoscale environments.
- Utilizing **Python** to develop analytical frameworks and conduct comprehensive **data analysis**, uncovering insights connecting physical mechanisms with anomalies in large-scale simulation results.

Indian Institute of Technology, Madras – Project Associate **Aug. 2020 – Apr. 2021**

Department of Aerospace Engineering *Chennai, India*

- Executed **Large-Eddy Simulations (LES)** of high Reynolds number channel flows with non-equilibrium effects.
- Implemented **wall-modeled LES (wall stress model)** and sensor-based computations for accurate transition flow modeling into high-fidelity **in-house CFD solvers**.

Havells India Ltd. – Research Engineer **July 2019 – Aug. 2020**

Center for Research and Innovation *Delhi, India*

- **Patented Venturi Air Cooler:** Designed an energy-efficient device eliminating water pumps by harnessing pressure/temperature profiles. Secondary inventor on patent **IN202011018315**.
- Developed analytical models leveraging **Blade Element theory** and **CFD** to design aerodynamic hardware, significantly improving volumetric flow rates and **minimizing power consumption** before scalable production.

University of Texas at Austin / IIT Delhi – Teaching Assistant **2017 – 2025**

Mechanical Engineering Departments *Austin, TX & Delhi, India*

- Instructed tutorial sessions and conducted automated evaluations using Python for Applied Thermodynamics (2025), Computational Fluid Dynamics (2022), Propulsion (2021), and Gas Dynamics (2018).

Publications & Patents

Publications: *Ozonated Monolayer Graphene for Extended Performance and Durability in Hydrogen Fuel Cell Electric Vehicles.* Devi Mutyala, Deji Akinwande, et al. ACS Nano (2025).

Publications: *Room-Temperature Pulsed Laser Deposition of Boron Nitride for Enhanced Fuel Cell Selectivity.* Devi Mutyala, Deji Akinwande, et al. ACS Nano (2025).

Patents: *Venturi Air Cooler.* Issued January 17, 2024. Patent Number IN202011018315.

Awards & Achievements

Academic Awards: Recipient of the **UT Austin Engineering Fellowship** (2021-2025) and the **IIT Kanpur Academic Excellence Award** awarded to the top 5% of the class (2014-2015).

Leadership and Service: Recipient of the **Institute Blues** (2016), the highest honor at IIT Kanpur for exceptional sports leadership and institutional impact as Team Captain and Deputy Contingent Leader.

Athletics Achievements: Attained 8 Inter IIT Athletics records and 11 Gold medals in track and field at IIT. Served as Captain of the Institute Athletics Team and Deputy Contingent Leader.